USN | | | | | | | | | | [22EE53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

POWER ELECTRONICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	What is the relation between the Break Over Voltage (V_{BO}) and the Break Down Voltage (V_{BR}).	L1	CO1	**02**
	b.	Differentiate the single phase half wave and full wave converters.	L1	CO2	**02**
	c.	What is the modulation index (M) in the sinusoidal PWM and explain its significance.	L1	CO3	**02**
	d.	Draw the diagram for Class D chopper.	L1	CO4	**02**
	e.	List out the various control techniques in AC Voltage Controllers.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Obtain the characteristics of IGBT.	L3	CO1	**05**
		OR			
	b.	Explore the principle of operation of MOSFET.	L2	CO1	**05**
3.	**a.**	Trace the waveform of a single phase full converter with firing angle 30 degree and extinction angle 190 degree. Find the RMS Value of the output if the input supply is 230 V, 50Hz.	L3	CO2	**05**
		OR			
	b.	Derive the RMS value of the output voltage for a three phase full converter.	L4	CO2	**05**
4.	**a.**	Obtain the output voltage waveform for the single phase full bridge inverter with a proper circuit diagram.	L3	CO3	**05**
		OR			
	b.	Explain the multiple pulse width modulation.	L2	CO3	**05**

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5.	**a.**	What is the principle of operation of buck boost Converter and draw the output waveforms.	L2	CO4	**05**
		OR			
	b.	Derive the output voltage of a step down chopper by clearly mentioning the circuit diagram and waveform.	L3	CO4	**05**
6.	**a.**	Deduce the waveform of output voltage and output current of a single phase full wave AC Voltage Controller with R load.	L4	CO5	**05**
		OR			
	b.	Distinguish the various three phase ac voltage controller configurations with a neat sketch of circuit diagram.	L2	CO5	**05**
7.	**a.**	Draw the circuit diagrams of single phase cyclo converter with midpoint and bridge configurations.	L2	CO6	**05**
		OR			
	b.	Explore the step down cycloconverter with a discontinuous current whose frequency is one fourth of the supply frequency.	L3	CO6	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Write a short note on the applications of all the power electronic converters.	L2	CO1	**10**
		OR			
	b.	Discuss the various parts in a power electronic system with a neat sketch.	L2	CO1	**10**
9.	**a.**	Derive the RMS current of a single phase half wave rectifier with R load with a neat sketch of circuit diagram and waveform.	L4	CO2	**10**
		OR			
	b.	Obtain the waveform of three phase full converter for firing angle 60 degree by clearly explaining the operation using a circuit diagram.	L3	CO2	**10**
10.	**a.**	Explore the CSI when compared to VSI.	L2	CO3	**10**
		OR			

	b.	Identify the output phase voltage V_c and the line voltage V_{ca} of the three phase bridge inverter in 180 degree mode by explaining the procedure with switching of switches in each mode.	L4	CO3	**10**
11.	**a.**	Demonstrate the operation of class E type of inverter with neat diagram and waveforms.	L2	CO4	**10**
		OR			
	b.	Discuss the purpose of cuk converter by explaining the operation of the same in detail.	L2	CO4	**10**
12.	**a.**	A single phase half wave ac voltage controller feeds a load of 25 ohms with an input voltage of 110V, 50Hz. Firing angle for both the thyristors is 30 degrees. Calculate; **i.** average value of output voltage and **ii.** load power **iii.** average output current. **iv.** power factor	L3	CO5	**10**
		OR			
	b.	Obtain the following waveform of a single phase full wave ac voltage controller with RL Load keeping low value of L **i.** Source Current **ii.** Voltage across thyristor 2 By explaining the operation in detail.	L3	CO5	**10**
13.	**a.**	Draw a schematic diagram of three phase to single phase cycloconverter by explaining the operation in brief.	L2	CO6	**10**
		OR			
	b.	Obtain the waveform of the cycloconverter to change the output frequency to thrice to the input frequency with good circuit diagram and necessary explanation.	L3	CO6	**10**